

HCSM-54

Vacuum Stability and the DME Stability Coefficient

The AGM Closed Form of the Ryu-Takayanagi Strip Integral, the Crossover Zero Moments, and the Discrete Analogue

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Abstract

We establish the vacuum stability sector of the Hodge Complex Standard Model. The DME (Dimensional Measurement Equilibrium) stability coefficient

$$\gamma = \frac{c_2}{I(2)} = 0.70283946799007245929257819650854315216110316215188 \dots$$

controls the width of the vacuum entropy well $S(\varepsilon)/S_0 = 1 - \gamma\varepsilon^2 + O(\varepsilon^4)$ around the equilibrium dimension $D = 4$, and the strict local maximum property $S''(0)/S_0 = -2\gamma < 0$ is the DME stability theorem. We prove three new results. (i) The Ryu-Takayanagi strip integral at $\alpha = 2$ has the exact closed form

$$I(2) = 1 - \frac{1}{2} \text{AGM}(1, \sqrt{2}),$$

verified at 100 digit precision, equivalently $I(2) = 1 - \sqrt{2} \pi^{3/2} / \Gamma(1/4)^2 = 1 - \pi / (2\sqrt{2} K(1/\sqrt{2}))$. (ii) The second derivative $c_2 = 2I''(2)$ has an exact closed form in π , $\Gamma(1/4)$, and Catalan's constant G . (iii) The Brillouin-zone moments of the three crossover-zero integrands on the 8×8 torus are exact integers

$$\langle (\lambda - 4)^2 \rangle = 4, \quad \langle \lambda(8 - \lambda) \rangle = 12, \quad \langle (\lambda - 4)^2 \lambda(8 - \lambda) \rangle = 28,$$

following the pattern $d \cdot (2^n - 1)$ for $n = 1, 2, 3$ with $d = 4$, and the mirror ratio $\langle (\lambda - 4)^2 \rangle / \langle \lambda(8 - \lambda) \rangle = 1/3 = 1/(d - 1)$ is the generation count denominator. A negative result is also established: the discrete analogue of $I(2)$ is *not* a Brillouin-zone moment of a crossover-vanishing integrand. The closest candidate, the spectral sum $\langle \sqrt{\lambda} / (1 + \lambda) \rangle_{\text{BZ}} = 0.40083787 \dots$, matches $I(2) = 0.40092988 \dots$ to 0.023%; the residual is not a framework constant, as confirmed by an exhaustive search over ~ 80 framework expressions. The HCSM-native derivation of $I(2)$ from the substrate therefore requires a construction beyond free Ω^1 BZ moments. We separate the derivation into substrate level theorems, a single disclosed physical identification, and open items. All numerical claims are verified in reproducible Python at machine precision.

Derivation status summary.

- DERIVED (substrate level and analytic): the AGM closed form $I(2) = 1 - \text{AGM}(1, \sqrt{2})/2$; the closed form of $c_2 = 2I''(2)$ in $\pi, \Gamma(1/4), G$; the closed form of $\gamma = c_2/I(2)$; the crossover-zero moments 4, 12, 28; the mirror ratio $1/3 = 1/(d - 1)$.

- **DISCLOSED IDENTIFICATION:** the DME construction $\gamma = c_2/I(2)$, i.e. the identification of the DME stability coefficient with the ratio of the second to first derivative of the strip integral at $\alpha = 2$. This is the framework’s dimensional-rigidity structure.
- **NEGATIVE RESULT:** the discrete analogue of $I(2)$ was investigated for (a) the S6 spectral sum, (b) the mirror and triple-zero integrand families, and (c) the strip entanglement entropy of the free Ω^1 Fermi sea. No BZ moment matches $I(2)$ exactly. The best candidate, $\langle \sqrt{\lambda}/(1+\lambda) \rangle_{\text{BZ}}$, is 0.023% low.
- **OPEN:** the HCSM-native derivation of $I(2)$ from the substrate, required for a full first-principles derivation of γ .

Contents

1	Introduction	3
1.1	The vacuum stability problem	3
1.2	Summary of results	3
1.3	Derivation status	4
1.4	Organization	5
2	Setup: the DME vacuum and the strip integral	5
2.1	The dimensional deviation	5
2.2	The Ryu–Takayanagi strip integral	5
2.3	The DME stability coefficient	5
3	The AGM closed form	6
3.1	The strip integral at $\alpha = 2$	6
3.2	The second derivative	6
3.3	The stability coefficient	6
4	Crossover-zero moments	7
4.1	The Brillouin-zone structure	7
4.2	The three crossover points	7
4.3	The crossover moments	7
5	The discrete analogue: negative result	8
5.1	The S6 spectral sum	8
5.2	The near-miss	8
5.3	What the negative result does and does not rule out	9
6	The DME stability theorem	9
7	Status: derived, disclosed identification, and open	10
7.1	Derived	10
7.2	Disclosed identification	10
7.3	Open	10
8	Falsifiable predictions	11
9	Conclusion	11

A Reproducible verification	12
B Web Crawler Metadata	13
B.1 Dublin Core	13
B.2 Schema.org JSON-LD	13
B.3 BibTeX for Automated Indexing	14

1 Introduction

1.1 The vacuum stability problem

The Hodge Complex Standard Model (HCSM) derives Standard Model structure from the spacetime dimension $d = 4$ through the discrete substrate $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ [1, 2, 3]. The DME (Dimensional Measurement Equilibrium) sector supplies the framework’s stability mechanism: it shows that $D = 4$ is a stable fixed point of the vacuum entropy, and it computes the exact stability coefficient γ that controls the width of the entropy well.

The stability coefficient enters the framework’s downstream predictions in three places. First, the EFT vacuum equation of state is $w = -1 - (1 - \gamma)^2 / (2\pi)$ [13], giving $w = -1.0140540788576648688\dots$, consistent with DESI DR2 at $\Delta\chi^2 = 0.22$. Second, the bulk viscosity is $\zeta H / \rho = (1 - \gamma)^2 / (6\pi)$. Third, the time quantum in the Motion and Time Theorems is $T_{\text{DME}} = 1 / (2\gamma)$ [11, 12].

In the prior WIN-era treatment the stability coefficient was stated as $\gamma = c_2 / I(2)$ with $I(2)$ the Ryu-Takayanagi strip integral at $\alpha = 2$ and $c_2 = 2I''(2)$. The strip integral is a continuum object; the framework’s strength is that it derives Standard Model structure from a discrete substrate, so a continuum import at the root of the entropy well construction is a genuine open item.

The present paper addresses this. It establishes three structural facts:

1. **Closed form of $I(2)$ and γ (Theorem 3.1, Theorem 3.3).** The strip integral at $\alpha = 2$ is the halved arithmetic–geometric mean of 1 and $\sqrt{2}$, subtracted from 1. The second derivative c_2 has a closed form in π , $\Gamma(1/4)$, and Catalan’s G . The stability coefficient is their exact ratio.
2. **Crossover-zero moments (Theorem 4.1, Corollary 4.3).** The Brillouin-zone moments of the three crossover-zero integrands on the torus are exact framework integers, and their mirror ratio is the generation-count denominator.
3. **Negative result (Section 5).** The discrete analogue of $I(2)$ is not a BZ moment of a crossover-vanishing integrand. The closest candidate, the S6 spectral sum, is 0.023% low; the residual is not a framework constant.

1.2 Summary of results

The present paper establishes the following.

1. (Theorem 3.1). The Ryu–Takayanagi strip integral at $\alpha = 2$ satisfies

$$I(2) = 1 - \frac{1}{2} \text{AGM}(1, \sqrt{2}).$$

Numerically $I(2) = 0.40092988263220389628003875385983806088639366839217\dots$, verified at 100-digit precision.

2. (Theorem 3.2). The second derivative $c_2 = 2I''(2)$ has the closed form

$$c_2 = 4 + B(-1/4, 1/2) \left[\frac{\pi + 4}{8} + \frac{(\pi + 4)^2}{128} + \frac{1}{4} + \frac{1 - G}{8} \right],$$

where $B(-1/4, 1/2) = -4\pi^{3/2}\sqrt{2}/\Gamma(1/4)^2$ and G is Catalan's constant.

3. (Theorem 3.3). The DME stability coefficient is

$$\gamma = \frac{c_2}{I(2)} = 0.70283946799007245929257819650854315216110316215188\dots$$

matching the prior WIN-era value to 6.7×10^{-18} .

4. (Theorem 4.1). The Brillouin-zone moments of the three crossover-zero integrands satisfy

$$\langle(\lambda - 4)^2\rangle_{\text{BZ}} = 4, \quad \langle\lambda(8 - \lambda)\rangle_{\text{BZ}} = 12, \quad \langle(\lambda - 4)^2\lambda(8 - \lambda)\rangle_{\text{BZ}} = 28,$$

exactly, following the pattern $d \cdot (2^n - 1)$ for $n = 1, 2, 3$.

5. (Corollary 4.3). The mirror ratio $\langle(\lambda - 4)^2\rangle/\langle\lambda(8 - \lambda)\rangle = 1/3 = 1/(d - 1)$ is the generation-count denominator.
6. (Theorem 5.1). The discrete analogue of $I(2)$ is not a Brillouin zone moment of a crossover-vanishing integrand. The closest candidate $\langle\sqrt{\lambda}/(1 + \lambda)\rangle_{\text{BZ}} = 0.40083787\dots$ is 0.023% below $I(2)$, and the residual 9.2×10^{-5} is not a framework constant.

1.3 Derivation status

The present paper separates its content into three explicit categories.

Derived (substrate-level and analytic).

- The AGM closed form of $I(2)$ (Theorem 3.1).
- The closed form of $c_2 = 2I''(2)$ (Theorem 3.2).
- The closed form of $\gamma = c_2/I(2)$ (Theorem 3.3).
- The crossover-zero moments 4, 12, 28 (Theorem 4.1).
- The mirror ratio $1/3 = 1/(d - 1)$ (Corollary 4.3).

Disclosed identification.

- The DME construction $\gamma = c_2/I(2)$: the identification of the DME stability coefficient with the ratio of the second derivative $c_2 = 2I''(2)$ to the value $I(2)$ of the Ryu–Takayanagi strip integral at $\alpha = 2$. This is the framework's dimensional-rigidity structure and is disclosed per the HCSM-06/07 pattern.

Open.

- The HCSM-native derivation of $I(2)$ from the substrate (Section 5).
- The physical mechanism by which c_2 enters at the second order of the entropy well.

1.4 Organization

Section 2 recalls the DME vacuum and the strip integral. Section 3 proves the AGM closed form. Section 4 proves the crossover-zero moments. Section 5 establishes the negative result on the discrete analogue. Section 6 states the DME stability theorem. Section 7 catalogues the derivation status. Section 8 lists falsifiable predictions. Appendix A provides reproducible Python verification. Appendix B gives structured bibliographic metadata.

2 Setup: the DME vacuum and the strip integral

2.1 The dimensional deviation

The substrate is the periodic square lattice $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ with $N = 64$ sites. The dimension of the vacuum manifold is $D = 4$. The dimensionless deviation from the equilibrium dimension is

$$\varepsilon = D - 4. \quad (1)$$

The DME entropy well is

$$\frac{S(\varepsilon)}{S_0} = 1 - \gamma\varepsilon^2 + O(\varepsilon^4), \quad (2)$$

with γ the DME stability coefficient. The sign of the second derivative at $\varepsilon = 0$ is -2γ , so $D = 4$ is a strict local maximum of the vacuum entropy.

2.2 The Ryu–Takayanagi strip integral

The strip integral at parameter α is

$$I(\alpha) = \int_1^\infty \left[\frac{u^{2\alpha-2}}{\sqrt{u^{2\alpha}-1}} - u^{\alpha-2} \right] du. \quad (3)$$

It has the closed form

$$I(\alpha) = \frac{1}{2\alpha} \left[B(a, \tfrac{1}{2}) - \frac{1}{a} \right], \quad a = -\frac{1}{2} + \frac{1}{2\alpha}. \quad (4)$$

At $\alpha = 2$ (the DME value $\alpha = D - 2 = 2$), this gives $a = -1/4$ and

$$I(2) = \frac{1}{4} B(-1/4, 1/2) + 1. \quad (5)$$

2.3 The DME stability coefficient

The DME stability coefficient is

$$\gamma = \frac{c_2}{I(2)}, \quad c_2 = 2I''(2), \quad (6)$$

where the prime denotes differentiation with respect to α . The numerical value is

$$\gamma = 0.70283946799007245929257819650854315216110316215188\dots \quad (7)$$

The identification (6) is the framework’s dimensional-rigidity structure. It is disclosed in the present paper per the HCSM-06/07 pattern, and the AGM/closed-form work below shows that both $I(2)$ and c_2 are specific transcendental constants rather than opaque integrals.

3 The AGM closed form

3.1 The strip integral at $\alpha = 2$

Theorem 3.1 (AGM identity). *The Ryu–Takayanagi strip integral at $\alpha = 2$ satisfies*

$$\boxed{I(2) = 1 - \frac{1}{2} \text{AGM}(1, \sqrt{2})} \quad (8)$$

where AGM denotes the arithmetic–geometric mean. Equivalently,

$$I(2) = 1 - \frac{\sqrt{2} \pi^{3/2}}{\Gamma(1/4)^2} = 1 - \frac{\pi}{2\sqrt{2} K(1/\sqrt{2})}, \quad (9)$$

with K the complete elliptic integral of the first kind.

Proof (sketch). From (5), $I(2) = 1 + B(-1/4, 1/2)/4$. The Beta function is evaluated by reflection:

$$B(-1/4, 1/2) = \frac{\Gamma(-1/4)\Gamma(1/2)}{\Gamma(1/4)} = -\frac{4\Gamma(3/4)\sqrt{\pi}}{\Gamma(1/4)}.$$

Using $\Gamma(1/4)\Gamma(3/4) = \pi/\sin(\pi/4) = \pi\sqrt{2}$, this gives

$$B(-1/4, 1/2) = -\frac{4\sqrt{2} \pi^{3/2}}{\Gamma(1/4)^2}.$$

The classical identity $\text{AGM}(1, \sqrt{2}) = \sqrt{2} \pi^{3/2}/\Gamma(1/4)^2$ then gives the result. The remaining equivalences follow from $K(1/\sqrt{2}) = \Gamma(1/4)^2/(4\sqrt{\pi})$. $\square$

3.2 The second derivative

Theorem 3.2 (Closed form of c_2). *The second derivative $c_2 = 2I''(2)$ has the closed form*

$$\boxed{c_2 = 4 + B(-1/4, 1/2) \left[\frac{\pi + 4}{8} + \frac{(\pi + 4)^2}{128} + \frac{1}{4} + \frac{1 - G}{8} \right]} \quad (10)$$

with $B(-1/4, 1/2)$ as above and G Catalan’s constant.

Proof (sketch). Differentiate (4) twice with respect to α . The result is expressible in terms of the Beta function $B(a, 1/2)$ at $a = -1/4$, its α -derivatives, and the polygamma combinations $\psi_1(1/4)$ and $\psi_1(-1/4)$. Using $\psi_1(1/4) = \pi^2 + 8G$ and the reflection $\psi_1(-1/4) = \pi^2 + 8G + 16(1 - G)$ yields the stated form. Verified at machine precision in Appendix A. $\square$

3.3 The stability coefficient

Theorem 3.3 (Closed form of γ). *The DME stability coefficient is*

$$\boxed{\gamma = \frac{c_2}{I(2)} = 0.70283946799007245929257819650854315216110316215188 \dots} \quad (11)$$

i.e., a specific transcendental constant expressible in π , $\Gamma(1/4)$, and G . It matches the prior WIN-era numerical value to 6.7×10^{-18} .

Proof. Direct from Theorem 3.1 and Theorem 3.2. $\square$

Remark 3.4 (From opaque integral to specific constant). The prior WIN-era derivation treated $I(2)$ as a numerically evaluated continuum integral. Theorem 3.1 converts it into a specific arithmetic constant, the halved AGM of 1 and $\sqrt{2}$. The identification $\gamma = c_2/I(2)$ is therefore not an identification with an opaque object but with a specific closed-form ratio.

4 Crossover-zero moments

4.1 The Brillouin-zone structure

The discrete 5-point Laplacian on the $L \times L$ torus has eigenvalues

$$\lambda_L(k_1, k_2) = 4 - 2 \cos(2\pi k_1/L) - 2 \cos(2\pi k_2/L), \quad k_1, k_2 \in \{0, \dots, L-1\}. \quad (12)$$

In the continuum limit $L \rightarrow \infty$, this becomes

$$\lambda(k_1, k_2) = 4 - 2 \cos k_1 - 2 \cos k_2, \quad (k_1, k_2) \in [0, 2\pi]^2, \quad (13)$$

with BZ average

$$\langle f \rangle_{\text{BZ}} = \int_{[0, 2\pi]^2} f(\lambda(k)) \frac{d^2 k}{(2\pi)^2}. \quad (14)$$

The BZ average is the continuum limit of the lattice mean $\frac{1}{L^2} \sum_k f(\lambda_L(k))$.

4.2 The three crossover points

The mirror symmetry $\lambda \leftrightarrow 8 - \lambda$ has three natural crossover points in the BZ:

- $\lambda = 0$ at the corner $\Gamma = (0, 0)$: the constant mode.
- $\lambda = 4$ at the two points $X = (\pi, 0), (0, \pi)$: the self-paired eigenvalue, which is the mirror-fixed point and carries the 14-mode multiplet on the discrete torus.
- $\lambda = 8$ at the corner $M = (\pi, \pi)$: the anti-constant mode.

An integrand is called *crossover-vanishing* if it vanishes at all three of these λ values.

4.3 The crossover moments

Theorem 4.1 (Crossover-zero moments). *The Brillouin-zone moments of the three crossover-zero integrands on the torus are exactly*

$$\boxed{\langle (\lambda - 4)^2 \rangle_{\text{BZ}} = 4, \quad \langle \lambda(8 - \lambda) \rangle_{\text{BZ}} = 12, \quad \langle (\lambda - 4)^2 \lambda(8 - \lambda) \rangle_{\text{BZ}} = 28.} \quad (15)$$

Proof. Each moment is a polynomial in λ . Using $\langle \lambda^n \rangle_{\text{BZ}} = \int_{[0, 2\pi]^2} (4 - 2 \cos k_1 - 2 \cos k_2)^n d^2 k / (2\pi)^2$ and the standard moments $\langle \lambda \rangle_{\text{BZ}} = 4$, $\langle \lambda^2 \rangle_{\text{BZ}} = 20$, $\langle \lambda^3 \rangle_{\text{BZ}} = 112$, $\langle \lambda^4 \rangle_{\text{BZ}} = 656$, the polynomial expansions give

$$\langle (\lambda - 4)^2 \rangle_{\text{BZ}} = \langle \lambda^2 \rangle - 8 \langle \lambda \rangle + 16 = 20 - 32 + 16 = 4, \quad (16)$$

$$\langle \lambda(8 - \lambda) \rangle_{\text{BZ}} = 8 \langle \lambda \rangle - \langle \lambda^2 \rangle = 32 - 20 = 12, \quad (17)$$

$$\langle (\lambda - 4)^2 \lambda(8 - \lambda) \rangle_{\text{BZ}} = \langle (20 - 8\lambda + \lambda^2)(8\lambda - \lambda^2) \rangle = 28. \quad (18)$$

The polynomial expansions are exact; the moments of λ follow from the standard Fourier averages. Verified in Appendix A. $\square$

Corollary 4.2 (Mersenne pattern). *The crossover-zero moments satisfy*

$$\langle(\lambda-4)^2\rangle_{\text{BZ}} = d \cdot (2^1 - 1) = 4, \quad \langle\lambda(8-\lambda)\rangle_{\text{BZ}} = d \cdot (2^2 - 1) = 12, \quad \langle(\lambda-4)^2\lambda(8-\lambda)\rangle_{\text{BZ}} = d \cdot (2^3 - 1) = 28, \quad (19)$$

with $d = 4$. The values 4, 12, 28 are $C(8, 2)$ -adjacent integers: 28 = $C(8, 2)$ is the Hodge lift $\Omega^1|_{\lambda=4}$ dimension, and $12 = N/2 - 20$ appears in the σ -line charge bookkeeping of HCSM-24.

Corollary 4.3 (Mirror ratio). *The ratio of the two lowest crossover-zero moments is*

$$\boxed{\frac{\langle(\lambda-4)^2\rangle_{\text{BZ}}}{\langle\lambda(8-\lambda)\rangle_{\text{BZ}}} = \frac{4}{12} = \frac{1}{3} = \frac{1}{d-1}}. \quad (20)$$

The right-hand side is the reciprocal generation count, $1/(d-1) = 1/3$.

Remark 4.4 (Structural significance). The mirror ratio (20) is exact, and its value $1/(d-1)$ is a framework constant. This is the first crossover-level identity that closes exactly. It shows that the crossover-zero integrand family is not merely a numerical coincidence: the family's lowest two moments are framework integers whose ratio is the generation count.

5 The discrete analogue: negative result

5.1 The S_6 spectral sum

The natural candidate for a discrete analogue of $I(2)$ is the Brillouin-zone average

$$S_6(L) = \frac{1}{L^2} \sum_{k_1, k_2=0}^{L-1} \frac{\sqrt{\lambda_L(k_1, k_2)}}{1 + \lambda_L(k_1, k_2)}, \quad (21)$$

with λ_L the torus Laplacian (12). The continuum limit is $S_6(\infty) = \langle\sqrt{\lambda}/(1+\lambda)\rangle_{\text{BZ}}$.

5.2 The near-miss

Numerically, $S_6(L)$ converges monotonically to $S_6(\infty)$:

L	$S_6(L)$	gap to $I(2)$
8	0.39767350	0.81%
16	0.40047818	0.11%
32	0.40079375	0.034%
64	0.40083238	0.024%
128	0.40083718	0.023%
256	0.40083778	0.023%
∞	0.40083787...	0.023%

At $L = 256$, $S_6 = 0.40083778$ versus $I(2) = 0.40092988$. The residual is 9.2×10^{-5} .

Theorem 5.1 (Negative result on the discrete analogue). *The discrete analogue of $I(2)$ is not a Brillouin-zone moment of a crossover-vanishing integrand. Specifically:*

1. The S_6 spectral sum (21) satisfies $S_6(\infty) = I(2) - 9.20 \times 10^{-5}$; the residual is 0.023% of $I(2)$.

2. An exhaustive search over ~ 80 framework expressions (powers of $1/N$, $1/(d+1)$, $1/\pi$, $1/(2\pi)$, $1/(4\pi)$, $1/(16\pi)$, $1/\Gamma(1/4)$, powers of $\text{AGM}(1, \sqrt{2})$, powers of $I(2)$, G , γ , c_2 , δ_{edge} , $\zeta H/\rho$, and their low-order products) finds no framework constant whose ratio to the residual is a small integer.
3. For the crossover-vanishing integrand families $\{(\lambda-4)^2/(1+\lambda)^p\}$ and $\{\lambda(8-\lambda)/(1+\lambda)^p\}$, no member has BZ average equal to $I(2)$ at integer p . The S6 integrand itself vanishes only at Γ , not at the self-paired point X , and is therefore not in the crossover-vanishing family.
4. The strip entanglement entropy of the free-fermion Ω^1 Fermi sea on the 8×8 torus scales as a volume law (c_{eff} growing with L), not the log-law of a 2D CFT.

Proof. Parts (1)–(4) are numerical: verified in Appendix A. Part (2) uses the exhaustive gap search over the framework-constant list. Part (4) uses the Peschel covariance-matrix method on the Ω^1 sector (128 modes) with the free-fermion ground state at half-filling. $\square$

Remark 5.2 (Interpretation). The crossover-zero integrands govern the framework’s discrete rational structure: their moments are the framework integers 4, 12, 28, and their lowest ratio is the generation count $1/3$. The strip integral $I(2)$ is a *transcendental* constant (AGM-based) that does not reduce to this rational structure. This is a genuine structural separation: the substrate’s discrete rational skeleton is not the source of the RT strip integral.

5.3 What the negative result does and does not rule out

The negative result rules out:

- The free-fermion Ω^1 Fermi sea strip entropy as the discrete analogue.
- The S6 spectral sum as an exact BZ moment.
- Any single-moment construction over the crossover-vanishing integrand family.

The negative result does *not* rule out:

- A discrete analogue that lives in a different sector (Ω^0 , Ω^2 , or the full 56-dimensional $\lambda = 4$ lift).
- A discrete analogue that involves the DNLS-interacting ground state rather than the free Fermi sea.
- A discrete analogue that requires a nontrivial state-preparation or region-selection, not just a BZ moment.
- A discrete analogue that only emerges in a different continuum limit.

These are the natural targets for the future HCSM-native derivation of γ .

6 The DME stability theorem

Theorem 6.1 (DME stability). *The equilibrium dimension $D = 4$ is a strict local maximum of the vacuum entropy. The second derivative at the equilibrium is*

$$\frac{S''(0)}{S_0} = -2\gamma = -1.405678935980144918585156393018\dots < 0. \quad (22)$$

The equilibrium is unique: no other integer D in the range $1 \leq D \leq 8$ gives a strict local maximum with the same numerical value of $S''(0)/S_0$.

Proof. The entropy well (2) has second derivative -2γ at $\varepsilon = 0$. The value of γ is established in Theorem 3.3. Numerical verification of the strict-local-maximum property and its uniqueness is in Appendix A. $\square$

Corollary 6.2 (Downstream predictions from γ). *The EFT vacuum equation of state is*

$$w = -1 - \frac{(1 - \gamma)^2}{2\pi} = -1.0140540788576648688 \dots, \quad (23)$$

and the bulk viscosity is

$$\frac{\zeta H}{\rho} = \frac{(1 - \gamma)^2}{6\pi} = 0.004684692952554956 \dots \quad (24)$$

Both are consistent with DESI DR2 at $\Delta\chi^2 = 0.22$.

7 Status: derived, disclosed identification, and open

7.1 Derived

1. The AGM closed form $I(2) = 1 - \text{AGM}(1, \sqrt{2})/2$ (Theorem 3.1).
2. The closed form of $c_2 = 2I''(2)$ in π , $\Gamma(1/4)$, and Catalan's G (Theorem 3.2).
3. The closed form of $\gamma = c_2/I(2)$ (Theorem 3.3).
4. The crossover-zero moments 4, 12, 28 (Theorem 4.1).
5. The Mersenne pattern $d \cdot (2^n - 1)$ (Corollary 4.2).
6. The mirror ratio $1/3 = 1/(d - 1)$ (Corollary 4.3).
7. The DME stability theorem (Theorem 6.1).
8. The negative result on the discrete analogue (Theorem 5.1).

7.2 Disclosed identification

1. The DME construction $\gamma = c_2/I(2)$: the identification of the DME stability coefficient with the ratio of the second derivative $c_2 = 2I''(2)$ to the value $I(2)$ of the Ryu–Takayanagi strip integral at $\alpha = 2$. This is the framework's dimensional-rigidity structure.

7.3 Open

1. The HCSM-native derivation of $I(2)$ from the substrate (Section 5).
2. The physical mechanism by which c_2 enters at the second order of the entropy well, rather than at a different order.
3. The finite- L corrections to the crossover-zero moments on the 8×8 torus.
4. The connection between the crossover-zero integrands and the σ -line structure of HCSM-24.

8 Falsifiable predictions

1. **Equation of state.** $w = -1 - (1 - \gamma)^2/(2\pi) = -1.014054\dots$, consistent with DESI DR2 at $\Delta\chi^2 = 0.22$ and testable at $> 3\sigma$ by DESI DR3 + Euclid.
2. **Bulk viscosity.** $\zeta H/\rho = 0.00468469\dots$, entering the framework's EFT predictions for the dark-energy density perturbations.
3. **Strict local maximum.** The framework predicts that any deviation $\varepsilon = D - 4$ is unstable to decay toward $\varepsilon = 0$. A detection of a stable equilibrium at $D \neq 4$ would falsify the DME construction.
4. **Negative prediction on the discrete analogue.** No BZ moment of a crossover-vanishing integrand on the discrete torus equals $I(2)$ exactly. A future identification of such a moment would falsify Theorem 5.1 and re-open the γ derivation.

9 Conclusion

We have established the vacuum stability sector of the Hodge Complex Standard Model. The DME stability coefficient $\gamma = c_2/I(2) = 0.70283946799007245929257819650854315216110316215188\dots$ controls the width of the entropy well around $D = 4$, and the strict-local-maximum property $S''(0)/S_0 = -2\gamma < 0$ is the DME stability theorem.

The AGM identity $I(2) = 1 - \text{AGM}(1, \sqrt{2})/2$ reduces the strip integral to a specific arithmetic constant. The closed form of $c_2 = 2I''(2)$ in π , $\Gamma(1/4)$, and Catalan's G completes the derivation of γ as a specific transcendental constant. The crossover-zero moments 4, 12, 28 on the discrete torus are exact framework integers whose lowest ratio $1/3 = 1/(d - 1)$ is the generation count.

A negative result is also established. The discrete analogue of $I(2)$ is not a Brillouin-zone moment of a crossover-vanishing integrand. The closest candidate, the S6 spectral sum, is 0.023% below $I(2)$, and the residual is not a framework constant, as confirmed by an exhaustive search over ~ 80 framework expressions. The HCSM-native derivation of $I(2)$ from the substrate therefore requires a construction beyond free Ω^1 BZ moments. This is stated explicitly as an open item.

The present paper follows the HCSM-06/07 pattern: it derives everything in its own domain that can be derived, discloses the single physical identification $\gamma = c_2/I(2)$ explicitly, and names the specific open item that a future HCSM-native construction must close. The framework's strength is that it has fewer empirical anchors than the Standard Model, not that it has none.

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A Reproducible verification

All numerical claims in this paper are verified at machine precision in IEEE-754 double precision (and at higher precision where symbolic evaluation is used). The verification script is provided in the HCSM companion repository at `12-VERIFICATION/HCSM-54_verification.py`. It verifies:

1. The AGM identity $I(2) = 1 - \text{AGM}(1, \sqrt{2})/2$ at 100-digit precision (Theorem 3.1).
2. The closed form of $c_2 = 2I''(2)$ (Theorem 3.2).
3. The closed form of $\gamma = c_2/I(2)$ to 6.7×10^{-18} relative error (Theorem 3.3).
4. The crossover-zero moments 4, 12, 28 via the standard λ moments (Theorem 4.1).
5. The mirror ratio 1/3 (Corollary 4.3).
6. The S6 spectral sum convergence and the 0.023% residual (Theorem 5.1).
7. The exhaustive gap search over ~ 80 framework expressions (Theorem 5.1).
8. The strip entanglement entropy of the free Ω^1 Fermi sea and the volume-law scaling (Theorem 5.1).

Running `12-VERIFICATION/run_all.py` reproduces every numerical claim in the HCSM paper stack.

B Web Crawler Metadata

B.1 Dublin Core

DC.title: HCSM-54: Vacuum Stability and the DME Stability Coefficient
DC.creator: Preschutti, Stanley
DC.creator.orcid: 0009-0004-5445-1744
DC.date.issued: 2026-09
DC.type: Text / Preprint / White Paper
DC.format: application/pdf; text/x-latex
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DC.publisher: Entropia Research Institute
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DC.subject: Hodge Complex Standard Model; HCSM; DME;
vacuum stability; stability coefficient; Ryu-Takayanagi strip integral;
AGM; arithmetic-geometric mean; crossover-zero moments;
Mersenne pattern; entropy well; dimensional measurement equilibrium
DC.description: We establish the vacuum stability sector of the Hodge
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 $\gamma = c2/I(2) = 0.70283946799007245929257819650854315216110316215188\dots$
The Ryu-Takayanagi strip integral at $\alpha = 2$ has the exact closed
form $I(2) = 1 - \text{AGM}(1, \sqrt{2})/2$, verified at 100 digits. The
second derivative $c2 = 2 I''(2)$ has a closed form in π , $\Gamma(1/4)$,
and Catalan's G . The crossover-zero moments on the discrete torus
are exactly 4, 12, 28, following the pattern $d * (2^n - 1)$ for $n = 1$,
2, 3, and the mirror ratio $1/3$ is the generation count. A negative
result is established: the discrete analogue of $I(2)$ is not a
Brillouin-zone moment of a crossover-vanishing integrand; the
closest candidate is 0.023% low, and the residual is not a framework
constant.
DC.identifier: HCSM-54-2026-09

B.2 Schema.org JSON-LD

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The Ryu-Takayanagi strip integral at  $\alpha = 2$  has the exact closed
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crossover-zero moments on the discrete torus are exactly 4, 12, 28,
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is established: the discrete analogue of  $I(2)$  is not a
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B.3 BibTeX for Automated Indexing

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    Complex Standard Model. The DME stability coefficient is
     $\gamma = c_2/I(2) = 0.70283946799007245929257819650854315216110316215188\dots$ 
    The Ryu-Takayanagi strip integral at  $\alpha = 2$  has the exact closed
    form  $I(2) = 1 - \text{AGM}(1, \sqrt{2})/2$ , verified at 100 digits. The
    crossover-zero moments on the discrete torus are exactly 4, 12, 28,
    and the mirror ratio  $1/3$  is the generation count.}
}

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